

FDTT 704: Hypothesis Testing

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Introduction

- A (parametric) hypothesis is a statement about one or more population parameters.
- A hypothesis test consist of:
 1. Two complementary hypothesis: the null hypothesis and the alternative hypothesis denoted H_0 and H_1 respectively.
 2. A decision rule that specifies for which sample values the null hypothesis is not rejected and for whcih the sample values the null hypothesis is rejected in favor of the alternative hypothesis.

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- The set of sample values for which H_0 is rejected is called the **rejection or critical region**
- The complement of the critical region is called the **acceptance region**
- A hypothesis that if true completely specifies the population distribution is called a **simple hypothesis**, one that does not is called a **composite hypothesis**

Steps in an Experiment

1. State the hypothesis: What is the null and alternative hypothesis
2. Test the null hypothesis at the desired p-value using the appropriate statistical tests
3. State your statistical conclusion regarding the null hypothesis
4. Provide an interpretation of your statistical conclusion

One sample t-test

- The one sample t-test is used to determine if the difference between a single sample mean and a known or estimated population mean is statistically significant
- Used when the population standard deviation is unknown
- Used also when the data are of at least interval or ratio scales.

Hypothesis

- The two-tailed hypothesis

$$H_0 : \mu = \mu_0; \quad H_1 : \mu \neq \mu_0$$

- One-tailed upper hypothesis

$$H_0 : \mu \leq \mu_0 \quad H_1 : \mu > \mu_0$$

- One-tailed lower hypothesis

$$H_0 : \mu \geq \mu_0 \quad H_1 : \mu < \mu_0$$

Sample Statistic

- If $X \sim N(\mu, \sigma^2)$ then $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$.
- Applying the standard normal transform we get the test statistic as:

$$t = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0, 1)$$

- Generally, however we do not know the population standard deviation σ
- Instead, we estimate it using the sample standard deviation s

Cont'd

- Introducing this extra level of uncertainty into the test statistic changes the distribution of the test statistic to a Student's t:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

Example 1

Suppose you want to see if patients with PTSD have higher than normal systolic blood pressure. You sample 25 patients and obtain a sample mean Systolic BP of 124.22 and a sample standard deviation of 23.7527 mm Hg. Using an alpha value of 0.05 is this observed mean greater than a 'normal' Systolic BP of 120 mm Hg?

Solution

- The hypothesis is $H_0 : \mu = 120$ $H_a : \mu > 120$
- The test statistic is:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{124.22 - 120}{23.7527/\sqrt{25}} = 0.8876$$

- The t-value from tables is $t_{0.05,24} = 1.7108$
- Since the calculated value is less than the tabulated value we fail to reject the null hypothesis.

Example 2

A company claims that the average waiting time for customers at its service center is 10 minutes. A researcher suspects that the actual waiting time is different from this claim. A random sample of 10 customers gives the following waiting times (minutes):

8,9,11,10,12,7,9,12,13,19

Test the claim at a 5% significance level.

Solution

- The hypothesis for the study is:

$$H_0 : \mu = 10, \quad H_1 : \mu \neq 10$$

- The sample mean and sample standard deviation is:

$$\bar{X} = \frac{\sum X_i}{n} = \frac{110}{10} = 11$$

$$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n - 1}} = 3.399$$

Cont'd

- The test statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{11 - 10}{3.399/\sqrt{10}} = 10.233$$

- The critical value $df = 9, \alpha = 0.05, t_{0.025,9} = \pm 2.262$
- Decision Rule: Reject H_0 if $t < -2.262$ or $t > 2.262$
- Since $t = 10.233$, we reject H_0 . There is sufficient evidence at the 5% significance level to conclude that the average waiting time differs from 10 minutes.

In R

```
1 # reading in the data
2 waiting_time <- c(8,9,11,10,12,7,9,12,13,19)
3 t.test(waiting_time,
4         mu = 10,
5         alternative = "two.sided",
6         conf.level = 0.95)
```

One Sample t-test

```
data: waiting_time
t = 0.93026, df = 9, p-value = 0.3765
alternative hypothesis: true mean is not equal to 10
95 percent confidence interval:
 8.568254 13.431746
sample estimates:
mean of x
11
```


Independent Sample t-test

- An independent samples t-test is a statistical test used to determine whether there is a significant difference between the means of two independent groups.
- It compares the average value of a quantitative variable from two separate populations.

Cont'd

- The independent sample t-test is appropriate when:
 - i. The dependent variable is **continuous**
 - ii. The independent variable has two independent groups.
 - iii. Observations in one group are not related to observations in the other group.

Hypothesis

- Suppose we compare two population means: μ_1 : Mean of population 1, μ_2 : Mean of Population 2
- The two population means are equal:

$$H_0 : \mu_1 = \mu_2 \quad \text{or} \quad H_0 : \mu_1 - \mu_2 = 0$$

- The alternative hypothesis:

$$H_1 : \mu_1 \neq \mu_2; H_1 : \mu_1 > \mu_2; H_1 : \mu_1 < \mu_2$$

Assumptions of Independent Sample t-tes

- Independence: Observations within each group are independent
- Normality: The data in each group should approximately follow a normal distribution
- Homogeneity of variance: The two groups should have approximately equal variances

Test statistic

- The test statistic is given by:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- The pooled standard deviation is:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$



Example

A researcher wants to compare examination scores between students taught using two different methods.

Data:

Method A 70,75,80,85,90

Method B 65,70,72,75,78

Test whether the average score differ at $\alpha = 0.05$

Solution

- The sample means

$$\bar{x}_1 = 80, \bar{x}_2 = 72, \bar{x}_1 - \bar{x}_2 = 80 - 72 = 8$$

- The sample variances $s_1^2 = 62.5, s_2^2 = 24.5$

- Pooled standard deviation

$$s_p^2 = \frac{(5 - 1)(62.5) + (5 - 1)(24.5)}{5 + 5 - 2} = 42.5 \Rightarrow s_p = \sqrt{42.5} = 6.519$$

Cont'd

- The test statistic

$$t = \frac{80 - 72}{6.595 \sqrt{\frac{1}{5} + \frac{1}{5}}} = 1.92$$

- The critical value is $t_{0.025,8} = 2.306$. The decision is $1.92 < 2.306$
- Therefore fail to reject H_0 . There is insufficient statistical evidence to conclude that the two teaching methods produce significantly different examination scores.

In R

```

1 method_A <- c(70,75,80,85,90)
2
3 method_B <- c(65,70,72,75,78)
4
5 t.test(method_A,
6         method_B,
7         var.equal = TRUE)

```

Two Sample t-test

```

data: method_A and method_B
t = 1.9179, df = 8, p-value = 0.09142
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1.619106 17.619106
sample estimates:
mean of x mean of y
      80      72

```


Paired Sample t-test

- A paired samples t-test is a statistical test used to determine whether there is a significant difference between two related measurements taken from the same subjects or matched subjects.
- It compares the mean difference between two observations.
- A paired t-test is used when:
 - i. The same subjects are measured twice
 - ii. The observations are naturally paired

Hypothesis

- let μ_d = population mean difference. The null hypothesis is there is no average change; $H_0 : \mu_d = 0$ and the alternative hypothesis is $H_1 : \mu_d \neq 0$; $H_1 : \mu_d > 0, \mu_d < 0$

Assumptions of Paired t-test

- Paired observations: Each value in one sample must correspond to exactly one value in the other sample.
- Continuous dependent variable
- Differences are approximately normally distributed: The individual measurements do not necessarily need to be normal.
- Independent pairs: Each pair should be independent from other pairs.

Formula for Paired t-test

- The test statistic is: $t = \frac{\bar{d} - \mu_d}{s_d / \sqrt{n}}$

where

- i. $\bar{d}$ mean of differences
- ii. μ_d hypothesized mean difference
- iii. s_d standard deviation of differences
- iv. n number of pairs

Worked Example

A researcher wants to determine whether a training program improves students' examination scores. Scores before and after training:

Student	Before	After
1	60	65
2	55	60
3	70	75
4	65	70
5	50	58

Test whether the training program significantly improves scores at 5% level of significance.

Solution

Student	Before	After	Difference
1	60	65	5
2	55	60	5
3	70	75	5
4	65	70	5
5	50	58	8

- The mean difference $\bar{d} = \frac{\sum d_i}{n} = 5.6$
- The standard deviation $s_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n-1}} = 1.342$

Cont'd

- The test statistic is $t = \frac{5.6 - 0}{1.342/\sqrt{5}} = 9.33$
- Decision $t_{0.05,4} = 2.132 < 9.33$.
- There is sufficient statistical evidence that the training program significantly improves students' examination scores.

In R

```
1 before <- c(60,55,70,65,50)
2
3 after <- c(65,60,75,70,58)
4
5 t.test(after,
6        before,
7        paired = TRUE,
8        alternative = "greater")
```

Paired t-test

data: after and before

t = 9.3333, df = 4, p-value = 0.0003668

alternative hypothesis: true mean difference is greater than 0

95 percent confidence interval:

4.320892 Inf

sample estimates:

mean difference

5.6

Chi-Square Test

- The Chi-square test is a hypothesis test used to determine whether there is a relationship between two categorical variables.
- The chi-square test is a hypothesis test used for categorical variables with nominal or ordinal measurement scale.
- The chi-square test checks whether the frequencies occurring in the sample differ significantly from the frequencies that one would expect.
- Thus, the observed frequencies are compared with the expected frequencies and their deviations are examined.

When to Use Chi-Square Test of Independence

- Both variables are categorical.
- Data are presented as frequencies (counts).
- Observations are independent.
- Each individual belongs to only one category of each variable.

Hypotheses

- There is no association between the two variables forms the null hypothesis
- The alternative hypothesis is that there is an association between the variables.

Assumptions

- Data are frequencies (counts).
- Categories are mutually exclusive.
- Observations are independent.
- Expected frequency in each cell is preferably at least 5.

Expected Frequencies

- Under null hypothesis the expected frequency for each cell is:

$$E_{ij} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

- The chi-square test statistic is:

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

Example

A researcher wants to determine whether gender is associated with employment status. Observed frequencies are given in table below

Gender	Employed	Unemployed	Total
Male	40	20	60
Female	30	30	60
Total	70	50	120

Test the association at the 5% significance level.

Solution

- The hypothesis is:

H_0 : Gender and employment status are independent.

H_1 : Gender and employment status are associated.

- The expected frequencies are:

$$E_{11} = 35, E_{12} = 25, E_{21} = 35, E_{22} = 25$$

Cont'd

- The $\chi^2 = 3.428$. The tabulated value is $\chi^2_{0.05,1} = 3.841$
- There is insufficient statistical evidence at the 5% significance level to conclude that gender and employment status are associated.

In R

```
1 employment <- matrix(c(40,20,  
2                        30,30),  
3                      nrow = 2,  
4                      byrow = TRUE)  
5  
6 rownames(employment) <- c("Male","Female")  
7 colnames(employment) <- c("Employed","Unemployed")  
8 chisq.test(employment)
```

Pearson's Chi-squared test with Yates' continuity correction

data: employment

X-squared = 2.7771, df = 1, p-value = 0.09562